

Q1: A transformer is used to light a 100W and 110v lamp from a 220v mains. If the main current is 0.5A. The efficiency of the transformer is:
a) 11% b) 50% c) 80% d) 90%

Q2: In an A.C circuit capacitive reactance is equal to inductive reactance. The phase difference between the current and voltage will be:
a) Zero b) $\pi/4$ c) $\pi/2$ d) π

Q3: How does the energy difference between the consecutive energy levels vary as the quantum number "n" Increases?
a) Increases b) decreases c) remains unchanged d) first increases & then decreases

Q4: Potential barrier of silicon at 30°C is 0.7 ev. Which of the following can represent the potential barrier of silicon at 100°C?
a) 0.5 ev b) 0.7 ev c) 0.9 ev d) 1.1 ev

Q5: If C and R denote capacity and resistance, the dimensions of CR are:
a) $[M^0L^0T^1]$ b) $[M^0L^2T^{-2}]$
c) $[M^0L^0T^{-2}]$ d) $[M^0L^0T^0]$

Q6: If L denotes the inductance of an inductor through which a current I is flowing, the dimensions of LI^2 are:
a) $[M^1L^2T^{-2}]$ b) Not expressed in [MLT]
c) $[M^1L^{-2}T^1]$ d) $[M^2L^2T^{-2}]$

Q7: The magnitude of vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ are respectively 12, 5 and 13 units and $(\vec{A} + \vec{B}) = \vec{C}$, then the angle between $\vec{A}$ & $\vec{B}$ is:
a) 0° b) π c) $\pi/2$ d) $\pi/4$

Q8: If a car at rest accelerates uniformly to a speed of 114 Km/hr in 20sec. It covers a distance of:
a) 20 cm b) 400 m c) 1440 cm d) 2980 cm

Q9: A man is throwing balls in air. He throws next ball when previous one is at highest point. If he throws each ball after 2 sec, then the height to which ball rises is ($g=10ms^{-2}$)
a) 10 m b) 15 m c) 20 m d) 30 m

Q10: A sphere is suspended by a thread of length "l". The minimum horizontal velocity that has to be imparted to the sphere for it to reach the height of the suspension:
a) gl b) 2gl c) $\sqrt{gl}$ d) $\sqrt{2gl}$

$$h = \frac{1}{2}gt^2$$

$$h = \frac{1}{2} \times 10 \times 2^2$$

$$h = 20m$$

$$\frac{6.3 \times 20}{3} = 42$$

Q11: The ratio of angular speeds of minute hand and hour hand of a watch is:

- a) 1:12 b) 6:1 c) 12:1 d) 1:6

Q12: On a planet "X", a man throws a 500g mass with a speed of 20ms^{-1} and catches it as it comes down 20sec later. The weight of the mass is:

- a) 1 N b) 10^4 N c) 10 N d) 5×10^3 N

Q13: A lift is moving up with acceleration equal to $1/5$ of that due to gravity. The apparent weight of the 60 Kg man standing in a lift is:

- a) 48 Kg b) 72 Kg c) 60 Kg d) 100 Kg

Q14: A car turns a corner on a slippery road at a constant speed of 12ms^{-1} . If the co-efficient of friction is 0.4, the minimum radius of the arc in meters in which the car turns is:

- a) 72 b) 36 c) 18 d) 9

Q15: A train weighing 10^4 N is running on a level track with uniform speed of 36Kmh^{-1} . The frictional force is 0.5Kg f per quintal. The power of the engine is ($g=10\text{ms}^{-2}$):

- a) 0.5 KW b) 5KW c) 50KW d) 500KW

Q16: A uniform chain has mass "M" and length "L". It is lying on a smooth horizontal table with half of its length hanging vertically down. The work done in pulling the chain up the table is:

- a) $\frac{Mgl}{4}$ b) $\frac{Mgl}{3}$ c) $\frac{Mgl}{8}$ d) $\frac{Mgl}{16}$

Q17: Newton's law of gravitation applies to:

- a) small bodies only b) planets only
c) both small and big bodies d) only valid for solar system

Q18: The time period of a simple pendulum inside a satellite orbiting Earth is:

- a) Zero b) Infinity c) 24 hrs. d) 12 hrs.

Q19: The angular momentum of the Earth revolving round the sun is proportional to R^n where R is the distance between Earth and sun. The value of n is:

- a) 0.5 b) 1.0 c) 1.5 d) 2.0

Q20: A body of mass 2 Kg and radius of gyration 0.5 m is rotating about an axis. If its angular speed is 2.0rad/sec , the angular momentum of the body (in $\text{Kgm}^2\text{s}^{-1}$) is:

- a) 0.5 b) 1.0 c) 2.0 d) 1.5

Q21: A wheel is rotating at 900 rpm about its axis. When the power is cut off, it comes to rest in 1 minute. The angular retardation (assumed to be uniform) in rad/sec^2 is:

- a) $\pi/2$ b) $\pi/4$ c) $\pi/6$ d) $\pi/8$

Q22: The bulk modulus of rubber is $9.1 \times 10^8 \text{Nm}^{-2}$. To what depth a rubber ball be taken in a lake so that its volume is decreased by 0.1 %:

- a) 1 km b) 25 m c) 100 m d) 200 m

Q23: Water rises to a height of 5 cm in a glass capillary tube. If the area of cross section of the tube is reduced to $(1/16)^{\text{th}}$ of the initial value, the water rises to a height of:

- a) 10 cm ~~b) 20 cm~~ c) 30 cm d) 40 cm

Q24: A spring when stretched up to 2 cm has a potential energy "u". If it is stretched up to 10 cm, the potential energy would be:

- a) $\frac{u}{25}$ b) $\frac{u}{5}$ c) 5u ~~d) 25u~~

Q25: A particle is executing simple harmonic motion with amplitude "a". When its K.E is equal to its P.E, its displacement from the mean position is:

- a) $a/2$ ~~b) $a/\sqrt{2}$~~ c) $a/\sqrt{3}$ d) $a/3$

Q26: Boiling water is changing into steam. Under this condition the specific heat of water is:

- a) zero b) one ~~c) infinite~~ d) less than one

Q27: A frictionless heat engine can be 100 % efficient if its exhaust temperature is:

- a) 0°C ~~b) 0 K~~ c) equal to its input temp. d) half of its input temp.

Q28: A well cut diamond appears bright because:

- a) it emits light b) it is radioactive
c) of total internal reflection d) of its high density

Q29: A bird in air looks at a fish below it and inside water. If " h_1 " is the height of bird above the surface of water and " h_2 " is the depth of fish below the surface of water. If refractive index of water with respect to air is μ , then the distance of the fish as observed by the bird is:

- a) $h_1 + h_2$ ~~b) $h_1 + h_2/\mu$~~ c) $\mu h_1 + h_2$ d) $\mu h_1 + \mu h_2$

Q30: The wave length of light diminishes μ times in the medium. A diver from inside water ($\mu=1.33$) looks at an object whose natural color is green. He sees the object as:

- a) green ~~b) blue~~ c) yellow ~~d) red~~

Q31: A converging lens has a focal length of 1 m. The minimum distance between a real object and its real image formed by this lens is:

- ~~a) 4 m~~ b) 2 m c) 1 m d) 0.5 m

Q32: If Young's double slit experiment is shifted from air to water. Assuming that water is still and clear, it can be predicted that the fringe pattern will:

- a) remain unchanged b) disappear
~~c) shrink~~ d) be enlarged

Q33: Two equal charges are separated by a distance "d". A third charge placed on a perpendicular bisector at "x" distance will experience maximum coulomb force when:

a) $x = \frac{d}{\sqrt{2}}$
 c) $x = \frac{d}{2\sqrt{2}}$

b) $x = \frac{d}{2}$
 d) $x = \frac{d}{2\sqrt{3}}$

Q34: The total flux from the surface of a cylinder of radius "R" and length "L" which is placed in a uniform electric field "E" parallel to cylinder axis is:

- a) $2\pi R^2 E$ b) $\pi R^2 E$ c) $(\pi R^2)/E$ d) Zero

Q35: A hollow metal sphere of radius 5 cm is charged such that the potential on its surface is 10v. The potential at a distance of 2 cm from the centre of the sphere is:

- a) Zero b) 10v c) 4v d) $\frac{10}{3}$ v

Q36: Eight mercury droplets each having radius 1 mm and a charge of 0.066 pc merge to form single bigger drop. The potential of single drop is:

- a) 2.4v b) 1.2v c) 3.6v d) 4.8v

Q37: Two capacitors each of capacity 1 μ f are charged to potential 10v and 6v respectively. They are joined together in parallels, the common potential established is:

- a) 16v b) 8v c) 4v d) 1v

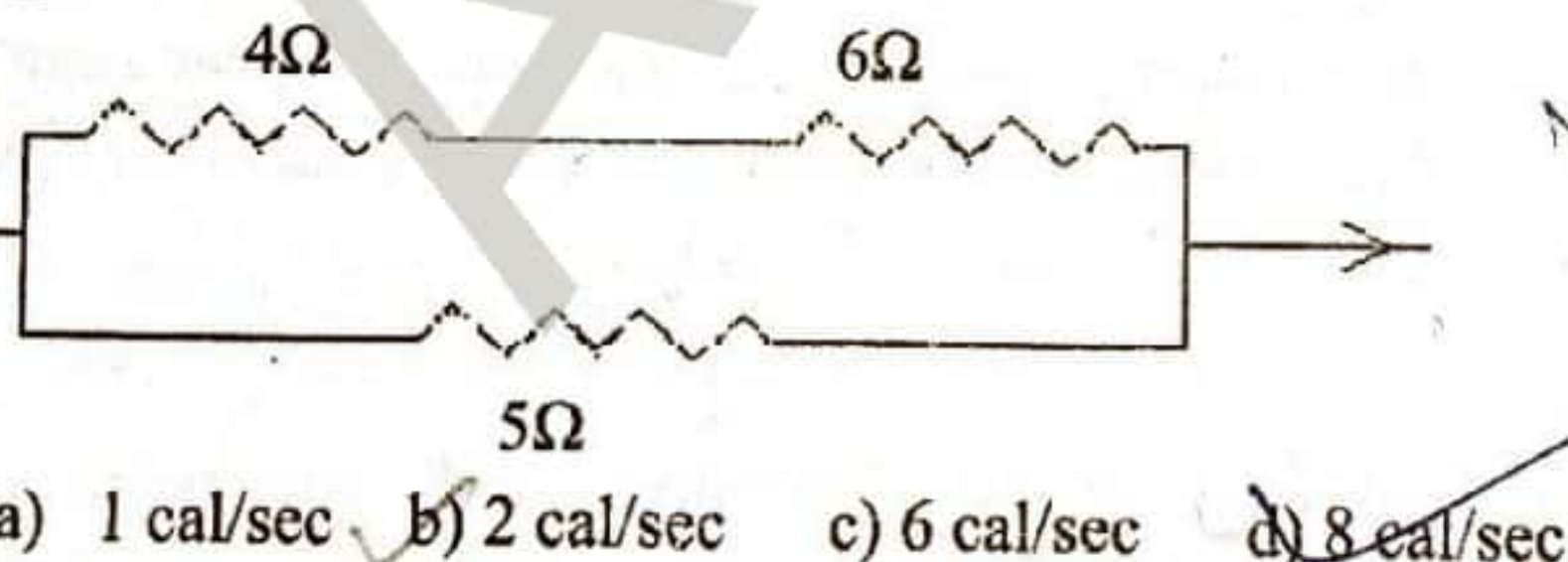
Q38: Two identical cells connected in series send 10A current through a 5 Ω resistance. When they are connected in parallels, they send 8A current through the same resistance. The internal resistance of cell is:

- a) 0 Ω b) 2.5 Ω c) 10 Ω d) 1 Ω

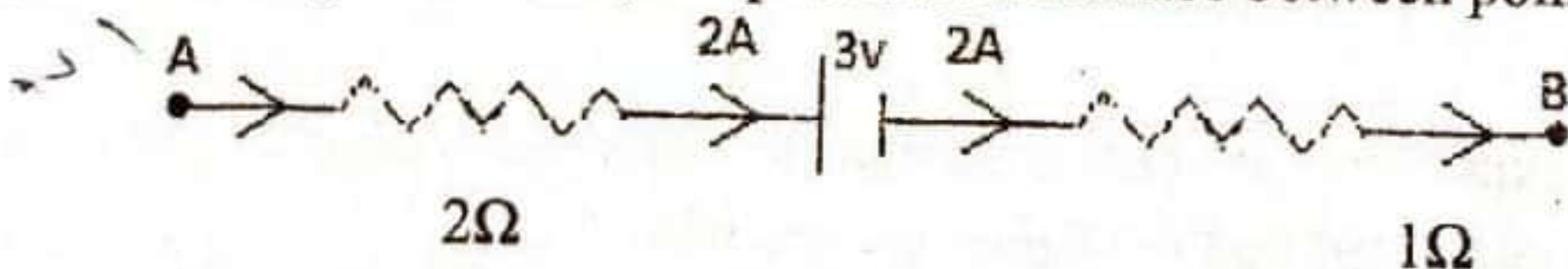
Q39: Masses of three wires of same metal are in the ratio of 1:2:3 and their lengths are in the ratio of 3:2:1. The electric resistances are in the ratio:

- a) 1:4:9 b) 9:4:1 c) 1:2:3 d) 27:6:1

Q40: In the circuit shown, the heat produced in the 5 Ω resistor is 10 cal/sec. The heat generated in 4 Ω resistor is:



Q41: In the given circuit, the potential difference between points A and B is:



Q42: A metallic rod carries direct current. Which of the following statements about the existence of magnetic field is true:

- a) it exists outside only
~~c) inside and outside~~
 b) it exists inside only
 d) neither inside nor outside

Q43: A circular coil of radius "R" carries a current "I". The magnetic field at its centre is "B". The distance from its centre at which the field becomes B/8 is:

- a) $\sqrt{2} R$
~~b) $\sqrt{3} R$~~
 c) $2R$
 d) $3R$

Q44: A magnetized straight wire of length "L" and dipole moment "P_m" is bent into a semi-circle. The dipole moment of the bent wire is:

- a) P_m
 b) P_m/π
 c) P_m/2π
~~d) 2P_m/π~~

Q45: Which of the following is used to determine the direction of induced current in a generator:

- a) Fleming's left hand rule
~~b) Fleming's right hand rule~~
 c) Maxwell's cork screw rule
 d) Ampere's swimming rule.

Handwritten calculations and diagrams:

- $\frac{14}{45}$
- $10 + 2 = 24F$
- $24 - 10 = 14$
- $R = 5\Omega$
- $V = 10$
- $q = 6 \times 10^3 \times 8^{\frac{2}{3}}$
- $6 \times 10^3 \times (2)^{\frac{3}{2}}$
- $6 \times 10^3 \times 4$
- 2.41
- $C = q/V$
- $10^{-3} = \frac{q}{V}$
- $10^{-3} = \frac{2.2}{V}$
- $10^{-3} = \frac{2.2}{V}$
- $24F = 16$
- $N =$
- $C_1 = 2.1N, C_2 = 2.0V$
- $10^{-3} = \frac{2.2}{V}$
- $10^{-3} = \frac{2.2}{V}$
- Diagram of a cylinder with radius R.
- Diagram of a semi-circular wire with radius R.
- Diagram of a circular wire with radius R.

CHEMISTRY

Q46: Aspirin is an acetylation product of:

- a) p - dihydroxybenzene
b) ☒ O-hydroxybenzoic acid
c) O-dihydroxybenzene
d) n - hydroxybenzoic acid

Q47: Polymerization of butadiene and styrene to give Buna -S is catalysed by:

- a) S
b) N
c) ☒ Na
d) Zeigler-Natta catalyst

Q48: Condensation product of caprolactum is:

- a) ☒ Nylon - 6
b) Nylon - 66
c) Nylon - 60
d) Nylon 6, 10

Q49: Among the following pairs of compounds, the one that illustrates the law of multiple proportions is:

- a) NH_3 and NF_3
b) ☒ SnCl_2 and SnCl_4
c) CS_2 and CO_2
d) NO_2 and NH_3

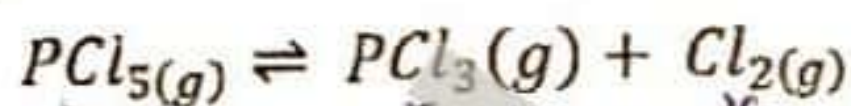
Q50: The maximum number of 5d electrons that can have spin quantum number $= +\frac{1}{2}$ are:

- a) 10
b) 2
c) 7
d) ☒ 5

Q51: The first emission line in the atomic spectrum of hydrogen in the Balmer series appears at:

- a) ☒ $\frac{5R}{36} \text{ cm}^{-1}$
b) $\frac{36}{5R} \text{ cm}^{-1}$
c) $\frac{36R}{7} \text{ cm}^{-1}$
d) $\frac{7}{36R} \text{ cm}^{-1}$

Q52: Phosphorus penta chloride dissociates as follows in a closed reaction vessel:



If total pressure at equilibrium of the reaction mixture is P and degree of dissociation of PCl_5 is ' x ', the partial pressure of $\text{PCl}_3(g)$ will be:

- a) $\left(\frac{x}{1+x}\right) P$
b) $\left(\frac{x}{1-x}\right) P$
c) $\left(\frac{2x}{1+x}\right) P$
d) $\left(\frac{x}{x-1}\right) P$

Q53: The pH of a solution of hydrochloric acid is 4, the molarity of solution is:

- a) 0.001
b) ☒ 0.0001
c) 4
d) 10^4

Q54: The solution of AgCl is unsaturated when:

- a) $K_{SP} = [\text{Ag}^+][\text{Cl}^-]$
b) $K_{SP} > [\text{Ag}^+][\text{Cl}^-]$
c) ☒ $K_{SP} < [\text{Ag}^+][\text{Cl}^-]$
d) none of these

Q55: The half life time of a reaction is constant if the order of the reaction is:

- a) ☒ zero
b) one
c) two
d) three

Q56: Which has the maximum freezing point:

- a) one molal NaCl aq. solution
b) one molal CaCl_2 aq. solution
c) ☒ one molal AlCl_3 aq. solution
d) one molal urea aq. solution

- Q57: Which of the following condition is not satisfied by an ideal solution?
- a) $\Delta H_{\text{mixing}} = 0$ b) $\Delta V_{\text{mixing}} = 0$
c) Raoult's law is obeyed (d) Formation of an azeotropic mixture

Q58: The heat of neutralization of strong acid and strong base is 57 kJ. The heat released when 0.5 mole of HNO_3 solution is added to 0.2 mole of NaOH solution is:

- a) 57 kJ b) 11.4 kJ c) 22.8 kJ d) 28.5 kJ

Q59: In an irreversible process $\Delta S_{\text{syst}} + \Delta S_{\text{surr}}$ is:

- a) < 0 b) ≥ 0 c) $= 0$ d) > 0

Q60: The standard reduction potential values of the elements A, B and C are +0.34V, -3.05V and +2.86V respectively. The order of their oxidizing power will be:

- a) $B > A > C$ b) $A < B < C$ c) $B < A < C$ d) $C < B < A$

Q61: Iron pipes lying under acidic soil are often attached to blocks of magnesium for protection from rusting. Magnesium offers protection to iron against corrosion because it:

- a) prevents air from reaching the surface of iron
b) forms corrosion resistant alloy with iron
c) its oxidation potential is lower than iron
d) is more readily converted into positive ions.

Q62: An electrochemical cell stops working after sometime because:

- a) electrode potential of both the electrodes becomes equal
b) electrode potential of both the electrodes becomes zero
c) anode electrode is completely eaten up
d) solution in the two half cells become neutral.

Q63: Which of the following is not a close packed arrangement?

- a) hcp b) ccp c) bcc d) fcc

Q64: Which of the following gas has the highest value of vander Waal's constant 'a'?

- a) Ne b) O_2 c) Cl_2 d) N_2

Q65: Colloidal solution of cellulose nitrate in alcohol is called:

- a) Aqua dag b) Argyrol
c) Purple of cassius d) Colloidion

Q66: Rate of physisorption increases with:

- a) decrease in temperature b) decrease in pressure
c) increase in temperature d) decrease in surface area

Q67: It is believed that atoms combine with each other such that the outermost shell acquires a stable configuration of 8 electrons. If stability was acquired with six electrons rather than 8; what will be the formula of stable fluoride ion:

- a) F^- b) F^+ c) F^{+2} d) F^{-2}

Q68: The number of naturally occurring p-block elements that are diamagnetic is:

- a) 7 b) 12 c) 6 d) 18

Q69: The number of bonding pairs (x) and lone pairs (y) around the central atom in the I_3^- ion are:

- a) $x = 2, y = 2$ **b) $x = 2, y = 3$** c) $x = 3, y = 2$ d) $x = 4, y = 3$

Q70: Bond order of O_2, O_2^+, O_2^{-1} and O_2^{-2} is in the order of:

- a) $O_2 < O_2^{-2} < O_2^{-1} < O_2^+$ b) $O_2^{-1} < O_2^{-2} < O_2^+ < O_2$
c) $O_2^{-2} < O_2^{-1} < O_2 < O_2^+$ d) $O_2^{-2} < O_2^{-1} < O_2^+ < O_2$

Q71: Which of the following molecule does not have a dipole moment:

- a) $CHCl_3$ b) IBr c) CH_2Cl_2 **d) BF_3**

Q72: Which of the following is preferentially formed on reaction of oxygen and potassium:

- a) K_2O_2 **b) KO_2** c) K_2O d) K_2O_3

Q73: Out of Si, Ge, Sn and Pb which forms stable ionic compounds in divalent state:

- a) Pb** b) Sn c) Ge d) Si

Q74: Hypohalites disproportionate in aqueous medium to give halides and halates. The rate of disproportionation increases in the order:

- a) $ClO^- < BrO^- < IO^-$** b) $IO^- < BrO^- < ClO^-$
 c) $BrO^- < ClO^- < IO^-$ d) $ClO^- < IO^- < BrO^-$

Q75: Which of the following compounds of manganese are amphoteric:

- a) MnO **b) Mn_2O_3** **c) Mn_2O_7** **d) none of these**

Q76: The colour of transition metal ions is due to absorption of some wavelength. This absorption is due to:

- a) d - p transition b) p - p transition
c) d - d transition d) f - f transition

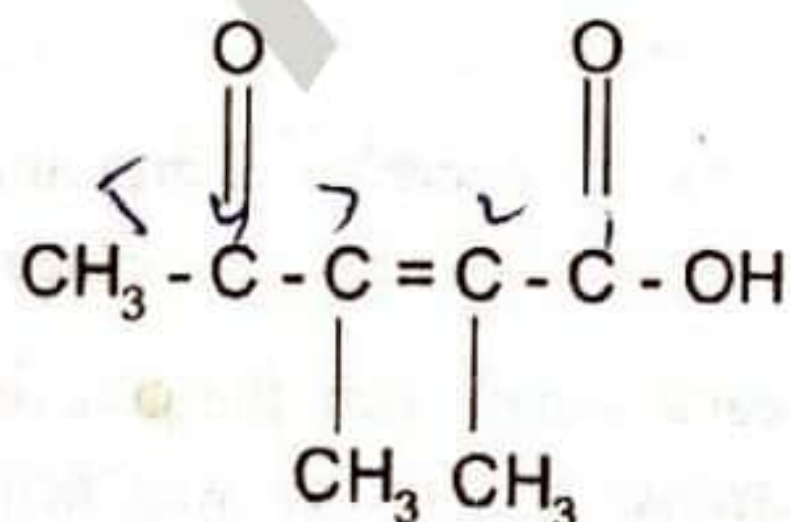
Q77: In which case racemic mixture is obtained on mixing its mirror images:

- a) cis - $[CoCl_2(en)_2]^+$** b) trans - $[CoCl_2(en)_2]^+$
 c) $Pt - [(NH_3)_2Cl_2]$ d) in all cases

Q78: Which of the following ligands form a chelate ring:

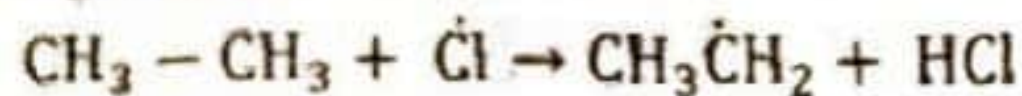
- a) Acetate **b) Oxalate** c) Cyanide d) Ammonia

Q79: The IUPAC name of the molecule



- a) 4-oxo-2,3-dimethylpent-2-en-1-oic acid
 b) 2-carboxy-3-methylpent-2-en-3-one
 c) 1,4-diketo-2,3-dimethylpent-2-en-1-ol
d) 2,3-dimethyl-4-oxo-pent-2-en-1-oic acid

Q80: Following step is:



- a) chain initiation ☒ b) chain propagation
c) chain termination ☐ d) chain disproportionation

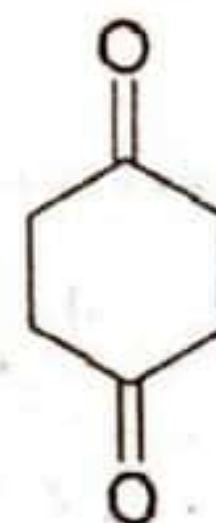
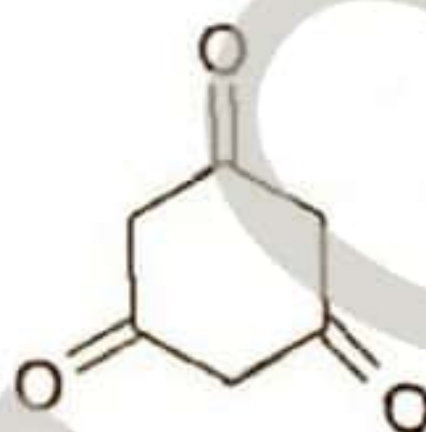
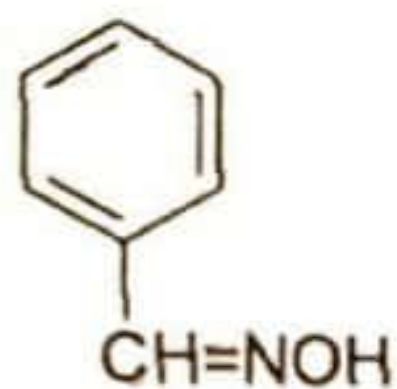
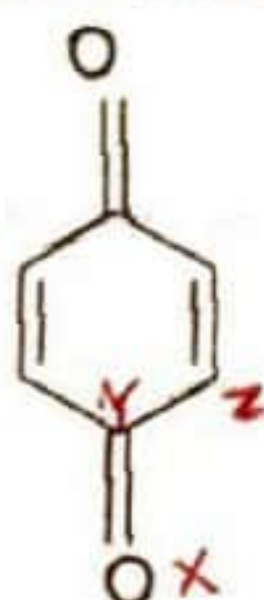
Q81: Inductive effect involves:

- a) displacement of sigma electrons ☒ b) delocalization of π -electrons
c) delocalization of σ -electrons ☐ d) displacement of π -electrons

Q82: Which of the following compounds can exist in optically active forms?

- a) Butan-1-ol ☐ b) Butan-2-ol ☒
c) Pentan-3-ol ☐ d) Heptan-4-ol ☐

Q83: Tautomerism is not exhibited by:



Q84: Methyl ketone group is identified by:

- a) Fehling's solution ☒ b) Schiff's reagent ☐
c) Iodoform test ☐ d) Tollen's reagent ☐

Q85: Ethanol when reacted with PCl_5 gives A, POCl_3 and HCl . 'A' reacts with silver nitrate to form 'B' (as major product) and AgCl . A and B respectively are:

- a) $\text{C}_2\text{H}_5\text{Cl}$ and $\text{C}_2\text{H}_5\text{OH}$ ☐ b) C_2H_6 and $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$ ☒
c) $\text{C}_2\text{H}_5\text{Cl}$ and $\text{C}_2\text{H}_5\text{NO}_2$ ☐ d) C_2H_6 and $\text{C}_2\text{H}_5\text{O}-\text{N}=\text{O}$ ☐

Q86: A new carbon-carbon bond formation is possible in:

- A) Cannizzaro's reaction ☒ B) Friedel-Craft's reaction ☒
C) Clemmensen reduction ☐ D) Reimer-Tiemann reaction ☒

- a) B and D only ☐ b) A and B only ☐
c) B and C only ☐ d) A, B and D only ☒

Q87: The most acidic among the following is:

- a) p-Cresol ☐ b) o-Cresol ☐
c) p-nitrophenol ☒ d) p-chlorophenol ☐

Q88: Hinsberg's reagent is:

- a/
b/
c) $\text{C}_6\text{H}_5\text{SO}_2\text{NH}_2$ ☐ d) $\text{CH}_3\text{CH}_2\text{COOC}_2\text{H}_5$ ☐

- Q89: Aniline is less basic than ethylamine. This is due to:
- Conjugation of lone pair of nitrogen with the ring
 - the insoluble nature of aniline
 - more K_b value of aniline
 - hydrogen bonding



- Q90: The process of conversion of sugar into alcohol and CO_2 by the action of yeast is called:
- fermentation
 - saponification
 - esterification
 - trans-esterification

$$\begin{aligned}
 & \frac{2.4 \times 10^{-4}}{4.8 \times 10^{-2}} \\
 & \frac{10}{48} \times \frac{121}{10} = \frac{24}{10} \times \frac{10}{48} \\
 & \quad \quad \quad (1.2 \times 10^{-2})^2 \\
 & \frac{24}{10} \times \frac{10}{48} = \frac{2.4 \times 10^{-4}}{4.8 \times 10^{-2}} - \frac{10^{-4} \times 10^2}{10^{-2}} \\
 & \quad \quad \quad 2 \times 10^{-2}
 \end{aligned}$$

BIOLOGY

- Q91: Phage mediated transfer of genetic material (DNA) from one bacterial cell to another is known as:
- ☒ a) Transformation
 - b) Conjugation
 - ☒ c) Transduction
 - d) All of the above
- Q92: Which of the following mineral ions is responsible for stomatal movements?
- a) Na^+
 - b) Ca^{2+}
 - c) Mg^{2+}
 - ☒ d) K^+
- Q93: Closure of leaflets of *Mimosa pudica* on touch is an example of:
- a) Tropic movement
 - b) Nutation movement
 - ☒ c) Siesmonastic movement
 - d) Thigmotropic movement
- Q94: First stable product in C-3 plants is called:
- ☒ a) Phosphoglyceric acid (PGA)
 - b) Oxaloacetic acid
 - c) Citric acid
 - d) Ribulose 1, 5, biphosphate (Rulp)
- Q95: The first genetically modified crop developed in India is:
- a) Flavr-Savr variety of tomato
 - b) Golden rice
 - ☒ c) Bt-cotton
 - d) Bt-corn
- Q96: Maggot is the larval form of
- ☒ a) House fly
 - b) Dragon fly
 - c) Mosquito
 - d) Honey bee
- Q97: Among the following who first saw and described a live cell:
- a) Robert Hooke
 - b) Robert Brown
 - ☒ c) Anton - Von Leeuwenhoek
 - d) Mathias - Schleiden
- Q98: Mitochondria are:
- ☒ a) The sites of Aerobic respiration
 - b) The sites for synthesis of Proteins
 - c) For Imparting specific colours to the plants
 - ☒ d) The sites to store nutrients

Q99: The insect (cockroach) is :

- a) Ureotelic
- b) Ammonotelic
- ☒ c) Uricotelic
- d) None of these

Q100: Which is the smallest known cell organelle

- a) Golgi complex
- ☒ b) Ribosome
- c) Lysosome
- d) Mitochondria

Q101: Which of the following is called the "Pacemaker of the heart" :

- ☒ a) Sino-atrial node (SAN)
- b) Atrio-Ventricular node (AVN)
- c) Bundle of His
- d) All of these

Q102: Sperm acrosome is derived from

- ☒ a) Golgi body
- b) ER
- ☒ c) Mysosome
- d) Mesosome

Q103: Abrin and Ricin are :

- a) Secondary metabolites of Alkaloids
- b) Secondary metabolites of Lectins
- c) Secondary metabolites of Essential oils
- ☒ d) Secondary metabolites of Toxins

Q104: The human beings having Klinefelter's syndrome have Chromosome composition

- a) $44 + XX0$
 - b) $44 + XXX$
 - ☒ c) $44 + XYY$
 - d) $44 + XXY$
- in males

Q105: What is the property known as when the binding of the chemical shuts off enzyme activity:

- a) Competitive Inhibition
- ☒ b) Inhibition
- c) Maximum velocity (V_{max})
- d) Enzyme substrate complex

Q106: In case of Earthworm a pair of male genital pores are present:

- ☒ a) On ventrolateral sides of the 18th segment
- b) At mid ventral line of 14th segment
- c) On ventrolateral sides of inter segmental grooves
- d) At the end of each body segment

Q107: Haemophilia is more common in males because it is a

- ☒ a) Recessive character carried by X-Chromosome
- b) Dominant character carried by Y-Chromosome
- c) Dominant trait carried by X-Chromosome
- d) Recessive trait carried by Y-Chromosome

Q108: Diffusion of gases (O_2 and CO_2) across alveolar membrane is one of the steps of:

- a) Digestion
- b) Circulation
- c) Both A and B
- ☒ d) Respiration

Q109: The glands which perform the excretory function in crustaceans like prawns are called:

- a) Flame cells
- b) Nephridia
- ☒ c) Antennal or Green glands
- d) Malpighian Tubules

Q110: Testicular hormones called Androgens are synthesized and secreted by:

- ☒ a) Leydig cells
- b) Sertoli Cells
- c) Scrotum
- d) Urthral meatus

Q111: Evolution for Darwin was gradual while De-Vries believed mutation caused speciation and hence called it:

- a) Natural selection
- b) Founder effect
- ☒ c) Saltation
- d) None of these

Q112: DNA fingerprinting involves identifying differences in some specific regions in 'DNA' sequence called:

- a) Satellite DNA
- ☒ b) Repetitive DNA
- c) SNP's
- ☒ d) VNTR

Q113: Which of the following is not a factor to affect "Hardy - Weinberg equilibrium :

- a) Gene mutation
- b) Genetic Drift
- c) Mutation
- ☒ d) Branching decent

Q114: Animals that have had their DNA manipulated to possess and express an extra (foreign) gene are known as:

- ☒ a) Transgenic animals
- b) Biological products
- c) Biopiracy
- d) Gene therapy

Q115: The spermatogonia present on the inside wall of seminiferous tubules multiply by:

- a) Meiotic division
- ☒ b) Mitotic division
- c) Fission
- d) Amitosis

Q116: Among the following who proposed central dogma in molecular biology :

- a) Robert Brown
- b) Bateson and Punnet
- ☒ c) Francis Crick
- d) Hans Kerb

Q117: Mammalian kidney is

- a) Pronephric b) Mesonephric ☒ c) Metanephric d) None of these

Q118: Making multiple identical copies of any template DNA is called:

- ☒ a) Replication
☒ b) Cloning
c) Gene transfer
d) All of these

Q119: Which of the following is placed in Prokaryote,

- ☒ a. Cyanophyceae → b. Chlorophyceae Green - algae
c. Phaeophyceae brown algae ☒ d. Rhodophyceae Red - algae

Q120: Casparian strips are found in roots of

- a. epidermal cells b. cortical cells ☒ c. endodermal cells d. pericycle

Q121: Which of the following is commonly called as pond silk

- ☒ a. Ulothrix ☒ b. Spirogyra fresh water Sphagnum
c. Chara d. Batrachospermum

Q122: Dry seeds when placed on moist filter paper, absorbs water and swells up due to

- a. Diffusion b. Osmotic diffusion
☒ c. Imbibition d. None of these

Q123: A Fern differs from Moss in having

- ☒ a. vascular tissue b. Independent gametophyte
c. swimming antherozoids d. distinct alternation of generation

Q124: The diffusion of solvent molecules into solution through semipermeable membrane is called as

- a. Imbibition b. Diffusion
☒ c. Osmosis d. All of these

Q125: The richest source of Gibberellin in higher plants is

- a. root b. stem
☒ c. leaf ☒ d. seed

Q126: Which of the following is not an auxin,

- a. IAA b. IBA
c. NAA ☒ d. Zeatin

Q127: The condition in which pollen is transferred from anther of a flower to stigma of another flower of the same plant is called

- a. autogamy ☒ b. geitonogamy
c. allogamy d. xenogamy

Q128: The term B.O.D is associated with pollution arising out of

- ☒ a. Water b. Air
c. Noise d. Land

Q129: Which of the following are included in biofertilizers

- a. Nitrogen fixing bacteria b. Nitrogen fixing Cyanobacteria
c. Mycorrhiza ☒ d. All of these

Q130: The interaction between members of two species in which member of one species capture, kill and eat up member of other species is called

- a. Allelopathy
- ☒ b. Predation
- c. Symbiosis
- d. Commensalism

Q131: The condition in which anther and stigma of a flower mature at the same time is

- a. cleistogamy
- ☒ c. dichogamy
- b. chasmogamy
- ☒ d. homogamy

Q132: Which of the following kingdoms includes unicellular eukaryotes?

- a. Monera
- b. Plantae
- c. Fungi
- ☒ d. Protista

Q133: Root hairs are present in which zone of tap root?

- a. Meristematic zone
- b. Root tip
- ☒ c. Maturation zone
- d. Elongation zone

Q134: Vascular bundles in the dicot stems are:

- ☒ a. Open and endarch
- b. Closed and endarch
- c. Open and exarch
- d. Closed and exarch

Q135: Which of the following cells in the pollen grain gives rise to male gametes?

- ☒ a. Generative cell
- ☒ b. Tube cell
- ☒ c. Vegetative cell
- d. None of the above

Sucrose → non-reducing
Maltose → reducing
Lactose → reducing

MATHEMATICS

Q136: If a line make angles α, β, γ with x -axis, y -axis and z -axis respectively then

$$\cos 2\alpha + \cos 2\beta + \cos 2\gamma = ?$$

a) 1

b) 2

☒ c) -1

d) -2

Q137: If $y = x^x$ then $\frac{dy}{dx} = ?$

☒ a) $x^x(1 + \log x)$

b) $x^x \log x$

c) $x \cdot x^{x-1}$

d) x^x

Q138: A is a matrix of order $n \times n$. Then inverse of A exists if and only if

a) A is symmetric

b) A is non-symmetric

☒ c) A is a singular matrix

☒ d) A is a non-singular matrix

Q139: $\frac{d}{dx}(\sec^{-1} x) = ?$

a) $\frac{1}{\sqrt{1-x^2}}$

b) $\frac{1}{\sqrt{x^2-1}}$

☒ c) $\frac{1}{|x| \sqrt{x^2-1}}$

d) $\frac{1}{1+x^2}$

Q140: B is a non-singular matrix of order $n \times n$ and I is identity matrix of order $n \times n$ then

$$|B| |I| =$$

a) $|B|$

b) B

☒ c) $B \text{ adj } B$

d) I

Q141: If x is a negative real number then argument $x =$

☒ a) π

b) $\pi/2$

☒ c) $-\pi/2$

d) 0

Q142: The angle between the vectors $8(i - k)$ and $i + j + k$ is

a) $\pi/4$

b) $\pi/3$

c) $\cos^{-1} \frac{1}{\sqrt{3}}$

☒ d) $\pi/2$

Q143: Number of terms in the expansion of $(a^2 + 6ab + 9b^2)^2$ is

a) 3

☒ b) 5

c) 4

d) 6

Q144: $\int \log x \, dx =$

a) $\frac{1}{|x|} + c$

☒ b) $x \log x - x + c$

c) $\frac{(\log x)^2}{2}$

d) none of these

c being constant of integration

Q145: If $f(x)$ is a real valued function of then :

☒ a) If $f(x)$ is continuous then $f(x)$ is also differentiable.

b) If $f(x)$ is differentiable then $f(x)$ is not continuous.

☒ c) If $f(x)$ is differentiable then it is also continuous.

d) None of these.

Q146: If $\cos x + \cos y + \cos z = 3$ then $\sin x + \sin y + \sin z =$

- a) 3 b) 1 c) 2 ~~d) 0~~

Q147: If A and B are feet of perpendiculars from the pt (2, 4, 5) to the x axis and y axis respectively, then the length of AB is

- ~~a) $2\sqrt{5}$~~ b) $\sqrt{29} - 1$ c) $\sqrt{41}$ d) $3\sqrt{5}$

Q148: If a plane meets co-ordinate axis at A, B and C such that the centroid of the triangle is $(\frac{1}{3}, \frac{2}{3}, \frac{4}{3})$ then the equation of the plain is

- a) $x + y + z = 3$ ~~b) $4x + 2y + z = 3$~~
~~c) $4x + 2y + z = 4$~~ d) $x + y + z = 9$

Q149: Equation of a circle is $x^2 + y^2 - 2x = 0$, its radius is equal to:

- ~~a) 1~~ b) 2 ~~c) 4~~ d) 3

Q150: Eccentricity of $16x^2 + 25y^2 = 400$ is

- ~~a) $\frac{4}{5}$~~ ~~b) $\frac{3}{5}$~~ ~~c) $\frac{5}{9}$~~ d) $\frac{1}{5}$

Q151: In statistical observations which of the following can not be negative
a) Mean b) Median c) Mode ~~d) Standard deviation~~

Q152: α, β are roots of the quadratic equation $x^2 + x + 1 = 0$ then $\alpha^5 + \beta^5 + 1 =$

- ~~a) 1~~ b) 2 ~~c) 0~~ d) -1

Q153: All possible 5 letter words (with or without meaning) are formed from the letters of 'SMALL'. The words are arranged in alphabetical order as in dictionary. The position of the word 'SMALL' will be

- a) 52nd ~~b) 58th~~ c) 59th ~~d) 46th~~

Q154: In three dimensional geometry $x = 0$ represents

- a) y-axis b) zx-plane c) XY-plane ~~d) YZ-plane~~

Q155: If $[x]$ stands for greatest integer function and n is a positive integer then

- $\int_1^n [x] dx =$
~~a) $\frac{n(n-1)}{2}$~~ b) $\frac{n(n+1)}{2}$ ~~c) $n+1$~~ d) n

Q156: The system equations $x + \lambda y - z = 0$, $\lambda x - y - z = 0$ and $x + y - \lambda z = 0$ have a non-trivial solution for

- a) Infinitely many values of λ b) for exactly one value λ
c) Exactly two values of λ ~~d) Exactly three values of λ~~

Q157: The angle between the lines $\frac{x-1}{1} = \frac{y-1}{1} = \frac{z-1}{2}$ and $\frac{x-1}{\sqrt{3}-1} = \frac{y-1}{-\sqrt{3}-1} = \frac{z-1}{4}$ is

- a) $\cos^{-1}(1/65)$ b) $\pi/6$ ☒ c) $\pi/3$ d) $\pi/4$

Q158: $\lim_{n \rightarrow 0} \frac{27^n - 9^n - 3^n - 1}{x^2} =$

- a) 12 ☒ b) 1 c) $\log 3$ d) $(\log 3)^2$

Q159: Z_1 and Z_2 are two complex numbers and $|Z_1 + Z_2| = |Z_1| + |Z_2|$ then $\arg z_1 - \arg z_2 =$

- ☒ a) 0 b) $\pi/2$ c) π d) $-\pi/2$

Q160: If $f(x) = \log x$ and $g(x) = e^x$ then $(f \circ g)x =$

- ☒ a) x b) e c) $e^x \log x$ d) 1

Q161: A and B are two events, $P(A) = 1/2$, $P(B) = 1/3$, $P(A \cap B) = 1/4$ then $P(\text{neither A nor B}) =$

- a) $1/7$ b) $7/12$ ☒ c) $5/12$ d) $1/12$

Q162: Every boy in a room shakes hands with every one else. The total number of hand shakes is 66. The number of boys in the room are

- a) 14 b) 13 ☒ c) 12 d) 11

Q163: If $\begin{bmatrix} -2 & 3 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -27 \\ 20 \end{bmatrix}$, then $(x, y) =$

- a) (14, 2) b) (13, 2) c) (15, 4) ☒ d) (15, 1)

Q164: The roots of the equation

$$\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} = 0$$
 are

- a) -1, -2 ☒ b) -1, 2 c) 1, -2 d) 1, 2

Q165: The slope of normal to the curve $xy = 12$ at the pt (3, 4) is

- ☒ a) $3/4$ b) $1/6$ c) $-4/3$ d) None of these

Q166: If $\int x \sin x \, dx = -x \cos x + K$, then $k = ?$

- a) $\cos x + c$ ☒ b) $\sin x + c$ c) $-\sin x + c$ d) $-\cos x + c$

Q167: $\int_0^1 \frac{(\sin^{-1} x)^2}{\sqrt{1-x^2}} dx = ?$

- a) $\frac{\pi^3}{8}$ b) $\frac{\pi^2}{24}$ ☒ c) $\frac{\pi^3}{24}$ d) $\frac{\pi^2}{12}$

Q168: The order and degree of differential equation $d^2y/dx^2 + (dy/dx)^{1/3} + x^{1/4} = 0$ are respectively

- a) 2, 6 ☒ b) 2, 3 c) 3, 3 d) 3, 4

Q169: The general solution of the differential equation $\log(dy/dx) = x + y$ is

- a) $e^x + e^y + c = 0$ b) $e^{-x} + e^y + c = 0$
c) $e^{-x} + e^{-y} + c = 0$ ☒ d) $e^x + e^{-y} + c = 0$

Q170: The integrating factor of $x \frac{dy}{dx} - 2y = x^2 + \sin 1/x^2$ is

- a) x^2 b) $-x^2$ ☒ c) $1/x^2$ d) $-1/x^2$

Q171: The max. value of $\frac{\log x}{x}$ in the interval $(2, \infty)$ is

- a) e b) $1/e$ c) $2e$ ☒ d) does not exist

Q172: The area bounded by $y = \sin x$ and the x -axis from $x = 0$ to $x = \pi$ is

- ☒ a) 2 b) π c) π^2 d) 2π

Q173: The area of triangle determined by vectors $3\hat{i} + 4\hat{j}$ and $-5\hat{i} + 7\hat{j}$ is

- a) $\frac{141}{2}$ square units ☒ b) $\frac{41}{2}$ square units
c) 41 square units d) None of these

Q174: $\hat{a}, \hat{b}, \hat{c}$ are three unit vectors, $\hat{a} + \hat{b} + \hat{c} = 0 \rightarrow$ then $\hat{a} \cdot \hat{b} + \hat{b} \cdot \hat{c} + \hat{c} \cdot \hat{a} = ?$

- ☒ a) 3 b) 0 c) -1 ☒ d) $-3/2$

Q175: The variance of statistical data x, y, z is 9. Then variance of $5x, 5y, 5z$ is

- a) 45 ☒ b) 225 c) $9/5$ d) $5/9$

Q176: The equation of line which is perpendicular bisector of the line segment joining $(-4, 6)$ and $(8, 8)$ is

- ☒ a) $6x + y - 19 = 0$ b) $6x - y - 19 = 0$
c) $6x + 2y - 19 = 0$ d) $x + 2y - 7 = 0$

Q177: The directrix of the parabola $y^2 = 16x$ is

- a) $y + 4 = 0$ ☒ b) $y - 4 = 0$ ☒ c) $x + 4 = 0$ d) $x - 4 = 0$

Q178: The general solution of the equation $\tan 3x = 1$ is

- a) $n\pi + \frac{\pi}{4}$ ☒ b) $\frac{n\pi}{3} + \frac{\pi}{12}$ c) $n\pi$ d) $n\pi \pm \frac{\pi}{4}$

(n being an integer)

Q179: Set A and Set B have 3 and 6 elements respectively. What can be the minimum number of elements in $A \cup B$?

- a) 3 b) 9 c) 18 ☒ d) 6

Q180: If ${}^{n+1}C_4 = 9 \cdot {}^nC_2$ then $n =$

- a) 9 b) 10 ☒ c) 11 d) 12

Handwritten notes and calculations:

Q179: Set A = {1, 2, 3}, Set B = {4, 5, 6, 7, 8, 9}. $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Minimum number of elements is 6.

Q180: ${}^{n+1}C_4 = 9 \cdot {}^nC_2$

$$\frac{(n+1)!}{4!(n-3)!} = 9 \cdot \frac{n!}{2!(n-2)!}$$

$$\frac{(n+1)n(n-1)(n-2)}{24} = 9 \cdot \frac{n(n-1)}{2}$$

$$(n+1)(n-2) = 108$$

$$n^2 - n - 2 = 108$$

$$n^2 - n - 110 = 0$$

$$(n-11)(n+10) = 0$$

$$n = 11$$

Q178: $\tan 3x = 1$

$$3x = \frac{\pi}{4} + n\pi$$

$$x = \frac{n\pi}{3} + \frac{\pi}{12}$$